

Different exposure levels of fine particulate matter and preterm birth: a meta-analysis based on cohort studies

Chenchen Liu¹ · Jiantao Sun¹ · Yuewei Liu² · Hui Liang³ · Minsheng Wang¹ · Chunhong Wang¹ · Tingming Shi²

Received: 3 February 2017 / Accepted: 23 May 2017 / Published online: 15 June 2017
© Springer-Verlag Berlin Heidelberg 2017

Abstract The previous studies estimated the association between PM_{2.5} (particulate matter with aerodynamic diameter less than or equal to 2.5 µm) exposure during pregnancy and preterm birth, only considered and highlighted the hazard effects of high levels of air pollutant exposure, and underestimated that low levels of pollutant exposure might also affect pregnancy outcome. We conducted a meta-analysis of 11 cohort studies, a total of more than 1,500,000 subjects. The results of these studies were pooled by exposure levels and study periods. PM_{2.5} exposure during pregnancy was positively associated with preterm birth (OR = 1.15, 95% CI = 1.07–1.23), and during the first trimester of pregnancy, low levels of PM_{2.5} exposure were also positively associated with preterm birth (OR = 1.17, 95% CI = 1.04–1.30). It is important to protect pregnant women from PM_{2.5} exposures, especially during their first trimester of pregnancy even when the ambient PM_{2.5} concentration is relatively low. More relevant health policy should be carried out to prevent hazard effect of air pollutants.

Keywords Exposure levels · Fine particulate matter · Preterm birth · Cohort study · Meta-analysis

Introduction

PM_{2.5} (particulate matter with aerodynamic diameter less than or equal to 2.5 µm) is a heterogeneous mixture of solid and liquid particles emitted from a variety of sources (Heo et al. 2014). Its small size, large specific surface area, and longer suspension time in the air provide the ability to cause adverse health effects, such as the development of chronic obstructive pulmonary disease (COPD) (DeVries et al. 2016) and coronary heart disease (Cormier et al. 2016). A large number of studies have indicated that ambient PM_{2.5} exposure may play an important role to the incidence of preterm birth (GK 2012; Nieuwenhuijsen et al. 2013), which has been associated with increased infant morbidity and mortality as well as potential increased morbidity in adulthood (GK 2012). It has been reported that reduction of air pollution in the first trimester during pregnancy is essential to maternal and neonatal health (Qian et al. 2016a, b). Recently, there is a growing number of studies that estimated the association between ambient PM_{2.5} exposure during pregnancy and preterm birth, but the results remain inconsistent and controversial (Fleischer et al. 2014; Poirier et al. 2015; Matte et al. 2016). For example, Poirier et al. reported that maternal PM_{2.5} exposure might be associated with preterm birth in an area of relatively low air pollution concentration (Poirier et al. 2015), while Fleischer et al. reported that outdoor PM_{2.5} concentrations were not associated with preterm birth (Fleischer et al. 2014).

Some systematic reviews or meta-analyses have been conducted to summarize the relationship between air pollution and preterm; however, there were notable limitations including (1) simply combined the results of articles with significant

Responsible editor: Philippe Garrigues

✉ Tingming Shi
tmingshi@hbcddc.com

✉ Chunhong Wang
wchunhong027@whu.edu.cn

¹ School of Health Sciences, Wuhan University, 115 Donghu Road, Wuhan, Hubei 430071, China

² Hubei Provincial Center for Disease Control and Prevention, 6 Zhuodaoquan North Road, Wuhan, Hubei 430079, China

³ Medical Research Center for Structural Biology, School of Basic Medical Sciences, Wuhan University, 115 Donghu Road, Wuhan, Hubei 430071, China

heterogeneity without testing the sources of it (Wang et al. 2009), (2) only considered and highlighted the adverse effect of high-level air pollutant exposures and underestimated the association between low-level pollutant exposures and abnormal pregnancy outcome (Chen et al. 2016), and (3) combined the results of a limited number of original studies with different research methods (Stieb et al. 2012; Zhu et al. 2014; Lamichhane et al. 2015).

Previous studies indicate that low-level exposure to ambient PM_{2.5} was associated with acute systemic inflammation and elevated mortality (Gent et al. 2003; Huttunen et al. 2012; Shi et al. 2015). Here we conducted a meta-analysis to provide summarized evidence on the associations between different levels of PM_{2.5} exposure and preterm birth. Cohort studies and nested case-control studies can provide a clear temporal sequence of exposure and disease, and help determine risk factors for contracting a new disease with reduced recall error. Results from long-term cohort studies are of substantially superior quality to those obtained from case-control/cross-sectional studies. Especially, prospective cohort studies are considered to yield the most reliable results in observational epidemiology. Therefore, in this study, we only included cohort or nested case-control studies, which provide evidence more persuaded (Xu 2008).

Materials and methods

Search strategy

We conducted a literature search in Embase, PubMed, Web of Science, Cochrane databases, and Wanfang database, identifying all published cohort studies that evaluated the association between PM_{2.5} exposure and preterm birth. Search terms included the following keywords: “PM_{2.5},” “fine particulate matter,” “air pollution,” “preterm,” “preterm birth,” “PTB,” “preterm delivery,” “premature birth,” “adverse pregnancy outcomes,” “adverse birth outcomes,” “cohort,” “longitudinal,” “perspective,” and “nested case-control.” We also manually searched reference lists of previous relevant published systematic reviews and meta-analysis. All original studies published in English or Chinese were considered.

Inclusion and exclusion criteria

Initial screening of the articles was based on titles and abstracts to exclude studies unrelated to PM_{2.5} exposure and preterm. Studies that passed the initial screening were further independently reviewed by two authors to assess whether they were eligible for this meta-analysis. All eligible studies included should meet the following criteria: (a) At least one of the following words: “cohort,” “prospective study,” “longitudinal study,” or “nested case-control” should be appeared in the

study. (b) The study should have a clear definition of preterm: born at less than 37 completed weeks of gestation. (c) The study should present the annual average exposure level of ambient PM_{2.5} and odds ratio (OR) with its 95% confidence interval (CI) or data that could be transformed into these results. (d) Potential confounders including maternal age, ethnicity, parity, education level, and infant sex should be adjusted. Exclusion criteria are the following: (a) Presented results could not be transformed into OR and its 95% CI in risk of preterm with per 1-unit increase in PM_{2.5}. (b) Exposure and health effect in first trimester or entire pregnancy were not mentioned. (c) Time-series studies, case reports, case series, and studies are available only in abstract form.

Data extraction

Data extraction from included studies was carefully performed by two independent authors to ensure the accuracy of the extracted information. The extracted data included first author’s surname, year of publication, country, study design, study period, birth population, exposure period, ORs and 95% CIs, and covariates adjusted in each study. As the studies employed different strategies on adjusting for gaseous air pollutants, results from single-pollutant models were used to potentially reduce the heterogeneity between studies in the analysis. Disagreements between the two authors were resolved by consensus. Finally, we employed Newcastle-Ottawa Scale to assess the quality of all included studies (Zeng et al. 2012).

Statistical analysis

Effect estimates of the included studies were divided into two parts: total estimates of pregnancy period (first trimester and entire pregnancy) and estimates based on different exposure levels. In the latter analysis, both the first trimester and the entire pregnancy period were divided into two groups: high-exposure-level group and low-exposure-level group according to the WHO Interim target-3 for PM_{2.5} (annual mean 15 µg/m³). Each group of ORs was converted to an exposure unit of 10 µg/m³ increase in PM_{2.5} exposure, which facilitates comparisons of ORs from different studies. We then conducted a set of meta-analyses, in accordance with Meta-analysis of Observational Studies in Epidemiology guidelines (Stroup 2000), to quantitatively estimate the associations between different levels of PM_{2.5} exposure and preterm birth risks. The heterogeneity between studies was evaluated using the chi-squared-based *Q* statistic and *I*² statistic. If a *P* value for the *Q* statistic >0.05 and *I*² <25%, an absence of heterogeneity among the included studies was suggested and the results of fixed-effect model were selected; otherwise, the random-effect model would be chosen (Stroup 2000; DerSimonian and Laird 2015). Sensitivity analysis or subgroup solutions were conducted to explore the possible sources of heterogeneity

if there was a statistically significant heterogeneity. Funnel plot was used to detect potential publication bias, and Egger's regression was applied to test the funnel plot symmetry. We used software Review Manager 5.2 and Stata version 11.0 to analyze the data. All statistical tests were two-sided, and a P value <0.05 was considered statistically significant.

Result

Descriptive summary of studies

With initial screen of the title and abstract of 1382 original studies identified in the literature search, 96 studies were selected as potentially eligible studies. We further excluded 78 studies that were not cohort nor nested case-control studies, or were unrelated to ambient $PM_{2.5}$ exposure. After reviewing full texts of the remaining 18 studies by either reviewer to insure eligibility for the inclusion, 11 studies with a total of more than 1,500,000 subjects were finally included in this meta-analysis. The process of study selection is presented in detail in Fig. 1. The characteristics of individual studies are summarized in Table 1. A total of 7 studies employed a retrospective cohort design using birth record data (Brauer et al. 2008; Lee et al. 2012; Pereira et al. 2013, 2014, 2016; Hannam et al. 2014; DeFranco et al. 2016), 3 were prospective studies (Gehring et al. 2011; Ha et al. 2014; Qian et al. 2016a), and 1 was nested case-control study (Ritz et al. 2007). Most studies were conducted in America with study period ranging

from 1996 to 2013. Almost all included studies used monitoring station to assess pollutants' exposure. The high-exposure-level group included 6 studies with reported average exposure level ranging from $15.6 \mu\text{g}/\text{m}^3$ (Lee et al. 2012) to $70.8 \mu\text{g}/\text{m}^3$ (Qian et al. 2016a). The low-exposure-level group included 5 studies with reported mean exposure level ranging from $5.1 \mu\text{g}/\text{m}^3$ (Brauer et al. 2008) to $13.2 \mu\text{g}/\text{m}^3$ (DeFranco et al. 2016). Almost all studies adjusted for maternal age, status, sex of infant, body mass index (BMI), race, education, and parity as a covariate, more than a half of studies adjusted for cigarette smoking, while less than half adjusted for season of conception. The average Newcastle-Ottawa quality score was 8.

Pooled effects

Forest plots presented the combined estimates of the risk of preterm with a $10 \mu\text{g}/\text{m}^3$ increase in $PM_{2.5}$ exposure based on the first trimester during pregnancy and entire pregnancy period. Table 2 presents a summary of pooled effect estimates.

Total

Figure 2 presents total estimates of the association between $PM_{2.5}$ exposure during the pregnancy and preterm birth. The results showed a significant increase of preterm birth risk associated with $PM_{2.5}$ exposure during the first trimester ($OR = 1.15$, 95% $CI = 1.05$ – 1.24). During the entire pregnancy, the combined OR was 1.15 (95% $CI = 0.99$ – 1.33) with high heterogeneity between included studies ($I^2 = 85\%$, $P < 0.001$).

By exposure level

In the group of high levels of $PM_{2.5}$ exposure during the first trimester during pregnancy, the heterogeneity of the included studies was estimated in a random-effect model where $I^2 = 38\%$ and $P = 0.18$, suggesting a moderate heterogeneity (Fig. 3). We did not find any significant effects of high levels of $PM_{2.5}$ exposure and preterm risk ($OR = 1.11$, 95% $CI = 0.94$ – 1.32). As for low-exposure-level group, we used random-effect model estimate ($I^2 = 44\%$, $P = 0.13$), which showed a significant increase of preterm birth risk associated with low-level $PM_{2.5}$ exposure ($OR = 1.17$, 95% $CI = 1.04$ – 1.30).

For the entire pregnancy, the combined OR s of the included studies in high- and low-exposure-level groups were estimated in fixed-effect model ($I^2 = 0\%$, $P = 0.41$) and random-effect model ($I^2 = 47\%$, $P = 0.13$), respectively. Both exposure groups showed significant associations between $PM_{2.5}$ exposure and preterm birth risk. The pooled OR s for $PM_{2.5}$ exposure in high-level group and low-level group were 1.06 (95%

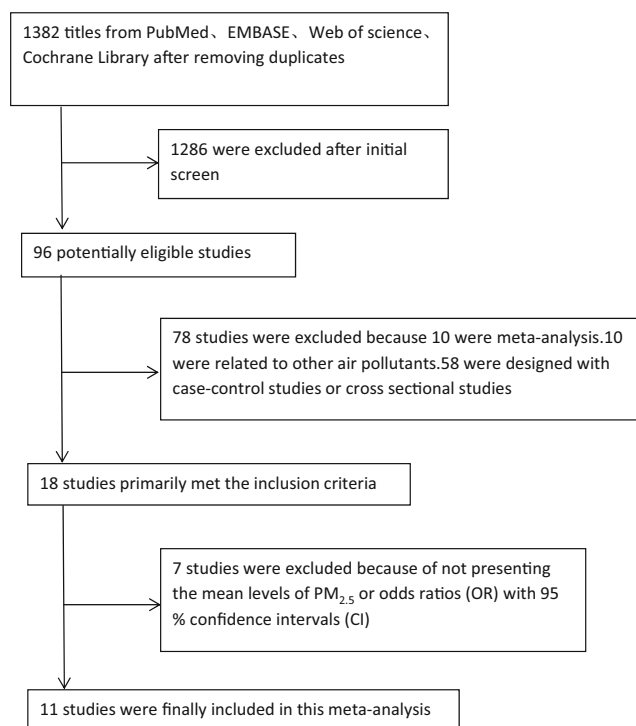


Fig. 1 Flowchart of the study selection process

Table 1 Characteristics of the studies included this meta-analysis

Author	Time period	Area	Sample size	Exposure period	Exposure level ($\mu\text{g}/\text{m}^3$)	Pooled OR (95% CI) ^a	Exposure assessment	Adjusted covariates	NOS
Pereira et al. (2016)	2008–2010	Rochester, USA	7121	First trimester	9.0	1.42 (1.14, 1.75)	MS	Age, education, Medicaid status, parity, sex of infant, BMI, race, education, parity, sex of infant	8
Qian et al. (2016a)	2011–2013	Wuhan, China	95,911	Entire pregnancy	70.8	1.06 (1.04, 1.08)	MS	Age, ethnicity, parity, education, job, address, sex of infant year of birth, season of conception	8
DeFranco et al. (2016)	2007–2010	OH, USA	224,921	First trimester	13.2	1.08 (0.88, 1.32)	MS	Age, race, education, parity, insurance, smoke, birth season and year, sex of infant	7
Hannam et al. (2014)	2004–2008	North West England	203,562	First trimester	22.1	1.00 (0.80, 1.25)	MS	Age, ethnicity, socioeconomic status, birth season, parity, BMI, smoke	8
Ha et al. (2014)	2004–2005	FL, USA	432,719	First trimester	9.6	1.10 (1.06, 1.15)	MS	Age, sex of infant, gestational age, education, race, status, prenatal care, smoke, alcohol	9
Pereira et al. (2014)	1997–2007	Perth, Australia	14,497	First trimester	8.5	1.00 (0.67, 1.49)	MS	season of conception, year of birth	8
Pereira et al. (2013)	2000–2006	CT, USA	61,688	Entire pregnancy	12.4	0.90 (0.60, 1.35)	MS	Age, race, parity, prenatal care insurance type, poverty, season of conception,	9
Lee et al. (2012)	1997–2002	Alleghe County, USA	34,705	First trimester	15.6	1.35 (1.00, 1.81)	MS	Age, smoke, parity	9
Gehring et al. (2011)	1996–1997	Netherlands	3853	Entire pregnancy	21.3	1.69 (1.04, 2.75)	MS	Age, race, parity, smoke, season of birth, and year of conception, race	8
Brauer et al. (2008)	1999–2002	Vancouver, Canada	70,249	Entire pregnancy	5.1	1.28 (0.98, 1.67)	MS	Age, BMI, parity, nationality, parental education, maternal smoking	9
Ritz et al. (2007)	2003	Los Angeles USA	58,316	First trimester	20.0	0.89 (0.63, 1.26)	LUR	Sex, ethnicity, parity, birth month and year, income, education, status	8
						0.76 (0.43, 1.34)	MS	Age, race, parity, education, and season of birth, smoke, alcohol, status	9

MS monitoring station, LUR land-use regression model

^a ORs were converted to an exposure unit of 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$

Table 2 Pooled associations between different exposure levels of PM_{2.5} (per 10 µg/m³ increment) during pregnancy and preterm birth risks

Groups	No. of studies	Heterogeneity test		Summary OR(95% CI)	Hypothesis test		<i>I</i> ² (%)	Egger's test	
		<i>Q</i>	<i>P</i>		<i>Z</i>	<i>P</i>		<i>t</i>	<i>P</i>
First trimester									
High-exposure group (≥15 µg/m ³)	4	4.83	0.18	1.11 (0.94, 1.32)	1.26	0.21	38%	−0.34	0.763
Low-exposure group (<15 µg/m ³)	5	7.09	0.13	1.17 (1.04, 1.30)	2.69	0.007	44%	1.03	0.380
Entire pregnancy									
High-exposure group (≥15 µg/m ³)	3	1.76	0.41	1.06 (1.04, 1.08)	5.94	<0.0001	0%	−4.16	0.150
Low-exposure group (<15 µg/m ³)	4	5.68	0.13	1.31 (1.06, 1.63)	2.44	0.01	47%	0.21	0.850

CI = 1.04–1.08) and 1.31 (95% CI = 1.06–1.63), respectively (Fig. 4).

Heterogeneity and publication bias

High heterogeneity was found between all included studies according to Fig. 2, while no significant heterogeneity was observed with consideration of exposure level (Table 2), indicating that exposure level was an important source of heterogeneity. In the entire pregnancy group, no heterogeneity was observed between studies included in high-exposure group ($I^2 = 0\%$, $P = 0.41$), and moderate heterogeneity was observed in low-exposure group ($I^2 = 47\%$, $P = 0.13$). For the first trimester group, moderate heterogeneities were observed between studies in high-exposure-level group ($I^2 = 38\%$,

$P = 0.18$) and low-level group ($I^2 = 44\%$, $P = 0.13$). Since no significant heterogeneities were observed in these four meta-analyses and no group of study number is more than 5, sensitivity analysis is inappropriate for this meta-analysis. The shape of the funnel plots seemed unsymmetrical in high-level group in the entire pregnancy, indicating the existence of publication bias (Fig. 5). Beyond that, we did not find any statistically significant publication bias in other groups. The funnel plots are shown in Fig. 5.

Discussion

PM_{2.5} has been a major public health concern globally due to its adverse health effects on reproductive health. This meta-analysis

Fig. 2 Forest plot for the pooled estimates of the risk of preterm with a 10 µg/m³ increase in PM_{2.5}

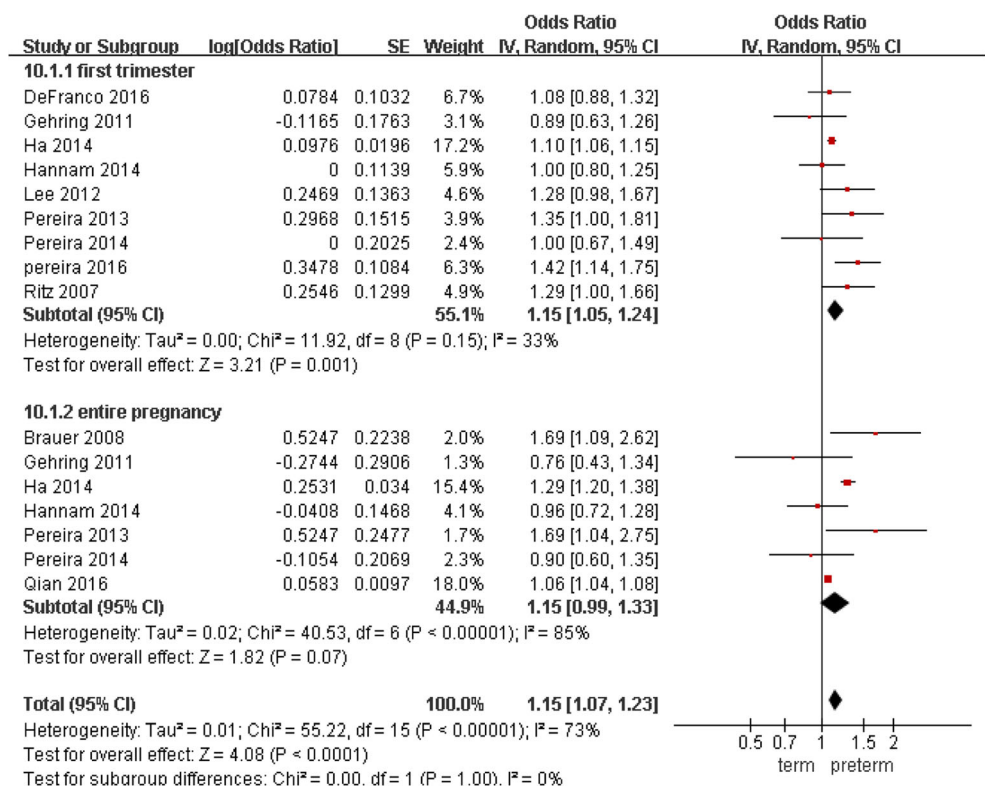
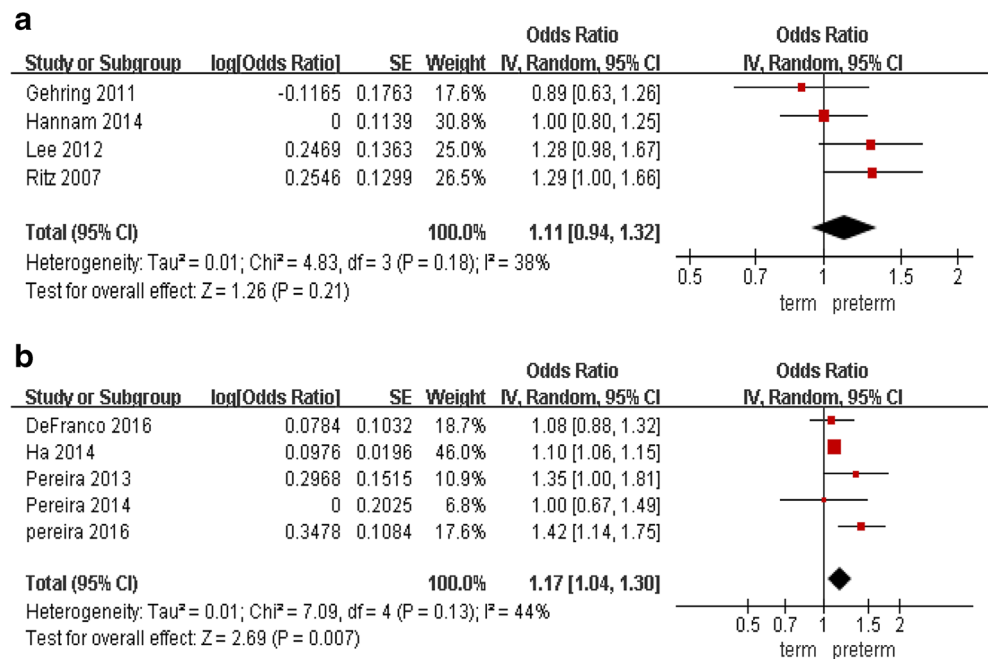


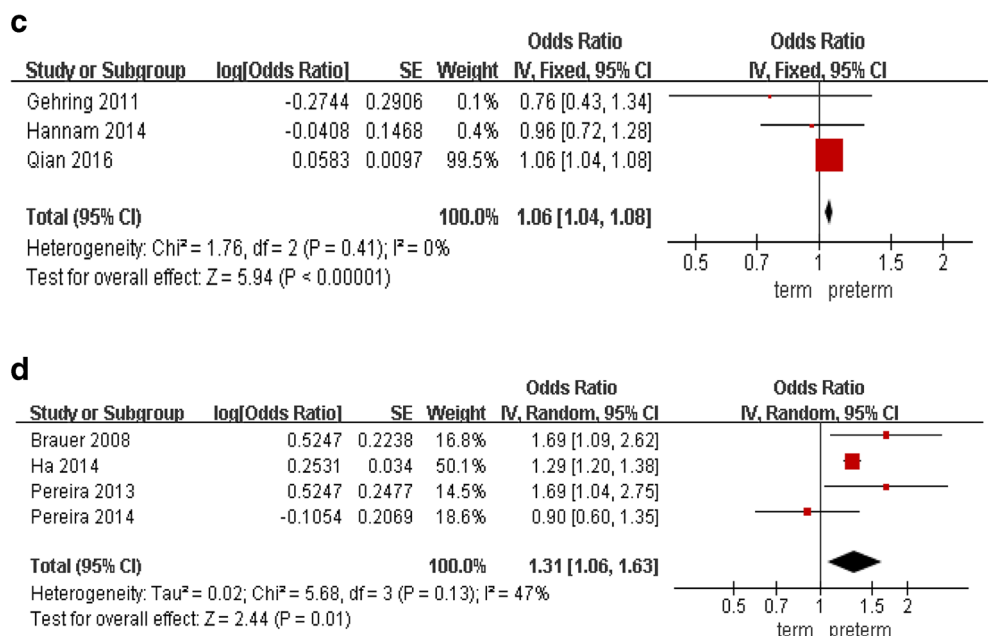
Fig. 3 Forest plot for the pooled estimates of the risk of preterm with PM_{2.5} exposure based on the first trimester of pregnancy: **a** high-level ($\geq 15 \mu\text{g}/\text{m}^3$) exposure and **b** low-level ($< 15 \mu\text{g}/\text{m}^3$) exposures



quantitatively assessed the association of different levels of PM_{2.5} exposure and preterm birth. We observed a positive but insignificant association between high-level PM_{2.5} exposure during the first trimester of pregnancy and preterm birth, but found a significant association between low-level PM_{2.5} exposure during the first trimester of pregnancy and preterm birth. This is inconsistent with the results reported by Sun et al., which showed no significant risks in preterm birth with a $10 \mu\text{g}/\text{m}^3$ increase in PM_{2.5} (Sun et al. 2015). The inconsistency may be due to the categories of the exposure level. Current epidemiologic evidence suggests that maternal PM exposure is correlated with several adverse

perinatal outcomes (Kannan et al. 2006). Women during the first trimester of pregnancy tend to reduce outdoor activities when the reported PM_{2.5} concentration is high, which may overestimate the ambient PM_{2.5} exposure and cause bias. But in the period when the reported PM_{2.5} concentration is relatively low, their consciousness of self-protection against air pollution is comparably weak, which may lead to the high risk of preterm birth. Therefore, our result highlighted the substantial potential to protect expectant mothers from PM_{2.5} exposure during the first trimester even the reported PM_{2.5} concentration is low. For the entire pregnancy period, we observed a clear significant

Fig. 4 Forest plot for the pooled estimates of the risk of preterm with PM_{2.5} exposure based on the entire pregnancy period: **c** high-levels ($\geq 15 \mu\text{g}/\text{m}^3$) exposure and **d** low-levels ($< 15 \mu\text{g}/\text{m}^3$) exposures



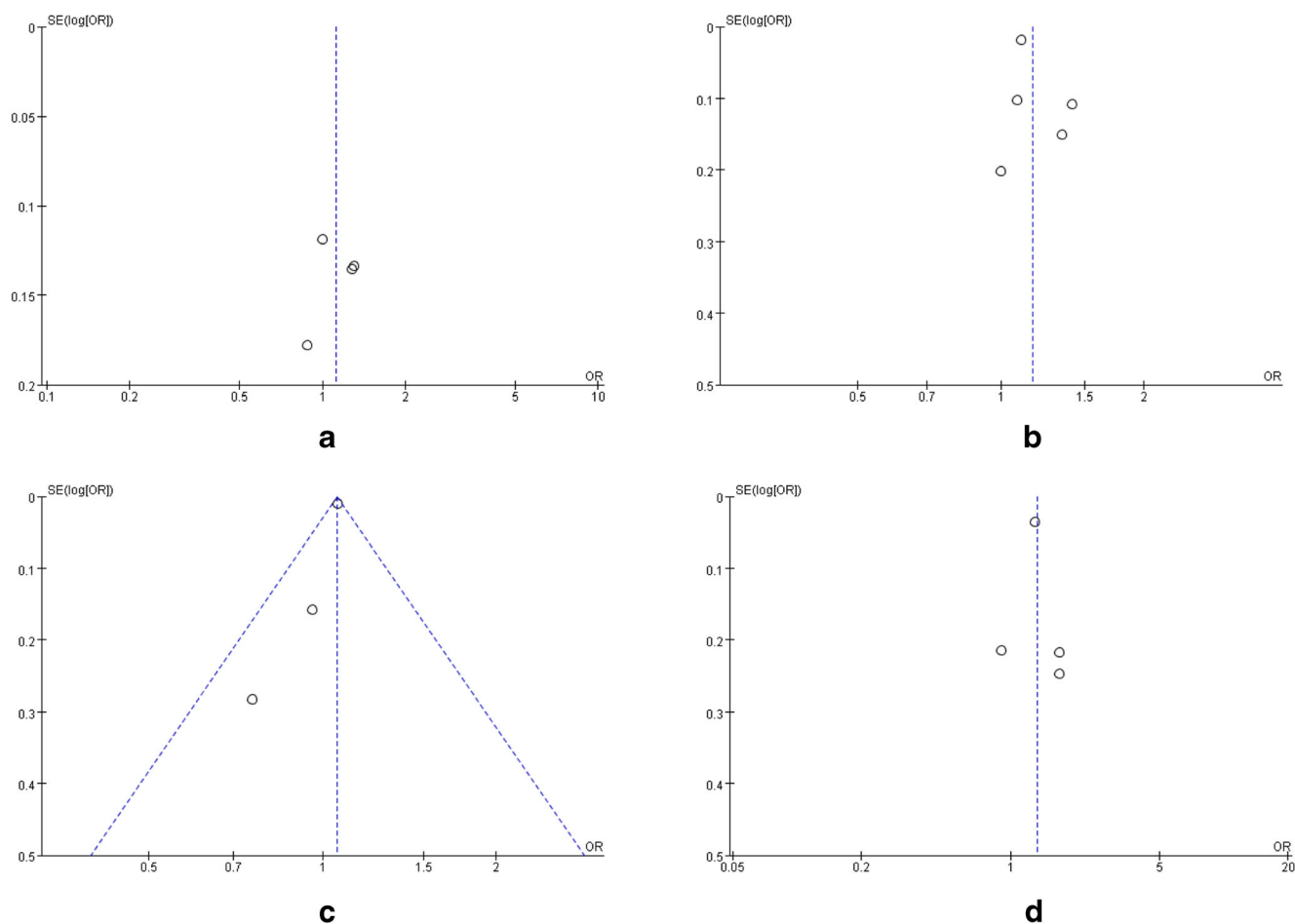


Fig. 5 Funnel plots for the meta-analyses: **a** high levels/first trimester, **b** low levels/first trimester, **c** high levels/entire pregnancy, and **d** low levels/entire pregnancy

association between both levels of $PM_{2.5}$ exposure and preterm birth, which is consistent with the conclusion of previous reviews (Maisonet et al. 2004; Šrám et al. 2005; Bosetti et al. 2010). Lamichhane et al. estimated a combined OR of 1.14 (95% CI = 1.06–1.22) for preterm birth per $10 \mu g/m^3$ increment in $PM_{2.5}$ exposure during the entire pregnancy (Lamichhane et al. 2015). Sapkota et al. also reported higher risk of preterm birth for each $10 \mu g/m^3$ increase in $PM_{2.5}$ during the entire pregnancy (OR = 1.15, 95% CI = 1.14–1.16) (Sapkota et al. 2010). These results further confirmed that exposure of $PM_{2.5}$ during the entire pregnancy period is an important risk factor for preterm birth.

As a syndrome caused by multiple pathologic processes, spontaneous preterm labor leads to 70% of preterm births (Romero et al. 2014a, b). Previous studies suggested that intra-amniotic infection, decidual senescence, and breakdown of maternal-fetal tolerance might be the main mechanisms of preterm birth (Cha et al. 2013; Romero et al. 2014a, b; Węgorzewska et al. 2014), and the adverse effects of air pollutant were also a unneglectable factor. Many previous meta-analyses have demonstrated the adverse effect of $PM_{2.5}$ exposure during pregnancy on preterm birth, but the conclusions remain controversial due to high heterogeneities between

included studies. According to many previous reviews, the imparity of study design, adjusted covariates, and the diversity of average ambient concentrations of pollutants consist of high percent of the heterogeneity (Glinianaia et al. 2004; Bosetti et al. 2010; Stieb et al. 2012; Sun et al. 2015), among which the study design weighted most. Cohort study was graded as level II in evidence grading system of evidence-based medicine, secondary to randomized controlled trial (Liu 2008). It has been one of the most important evidence for assessment of hazard exposure to human health. In many previous reviews (Bosetti et al. 2010; Zhu et al. 2014), the effect sizes of cohort studies were often combined with case-control studies and even ecological studies, which was somewhat unreasonable because the acquisition and analysis of data, the research process, and the solutions of probable bias were different among various study designs. Therefore, the studies included in this meta-analysis all employed cohort study design or nested case-control study design, which might prominently decrease heterogeneity between studies.

It is well known that different concentrations of pollutants result in different toxicity and health impacts, and the hazard

effects of low levels of air pollutant exposure to human health are often underestimated. According to all included 11 studies in this meta-analysis, the annual average concentrations of PM_{2.5} ranged from 5.1 µg/m³ (Vancouver, Canada) to 70.8 µg/m³ (Wuhan, China). Therefore, we divided our assessment into four parts based on the exposure levels and period: (a) high-level PM_{2.5} exposure during the first trimester of pregnancy, (b) low-level PM_{2.5} exposure during the first trimester of pregnancy, (c) high-level PM_{2.5} exposure during the entire pregnancy, and (d) low-level PM_{2.5} exposure during the entire pregnancy. Owing to the category of exposure levels, the heterogeneities of the included studies were insignificant. However, due to the small quantity of studies in each group, the capacity of conclusion explanation was still limited. More prospected studies with clear exposure levels are still warranted in future.

High heterogeneity among studies may be the main reason for inconclusive estimates. Heterogeneity is potentially an important difference in the results of individual studies being considered for a meta-analysis, the sources of which could be the disparity of subject chosen, study setting, and methodology (Hatala 2005). We controlled the study design, adjusted covariates, and quality evaluation rigorously during the progress of study selection. We also exclude studies with high heterogeneity in the statistical analysis (Jalaludin et al. 2007; Matte et al. 2016; Laurent et al. 2016). However, the results showed that although study designs, exposure levels, and main confounders partially explained the heterogeneity, moderate heterogeneities were still found in three of our analyses. This finding shows that heterogeneity may also be influenced by other confounders that we did not consider in this meta-analysis. The hazard of PM_{2.5} may vary by different geographic areas as well as its constituents. Therefore, the adverse effects on pregnancy outcome by PM_{2.5} exposure may vary by geographical locations and the main constituents of PM_{2.5}. The results of study conducted in Wuhan, China (Qian et al. 2016a), presented the serious air pollution (the annual average ambient concentrations of PM_{2.5} reach to 70.8 µg/m³), and the main constituents of PM_{2.5} are polycyclic aromatic hydrocarbons (PAHs) and water-soluble inorganic ions (Zhang et al. 2015), which were different from the results of other included studies conducted in USA or Europe. But the limited number of studies restricted us from conducting subgroup meta-analyses between studies based on different geographic areas and PM_{2.5} constituents. Although maternal age, status, parity, sex of infant, BMI, race, and education were adjusted in all included studies, there may still be other confounders that limited the ability to control, such as alcohol, season of conception, nutrition, maternal stress, and insurance. This could partly explain the heterogeneity. There have been some studies that used personal monitors to assess maternal exposure to air pollutants during pregnancy, which provided more precise results (Jedrychowski et al. 2004, 2009). So more studies with individual information would be benefit to future analyses.

Despite that all eligible studies were included as possible, we found evidence of publication bias based on Egger's test and funnel plot asymmetry for high-level PM_{2.5} exposure during the entire pregnancy and preterm birth. The restriction of languages (only studies published in English or Chinese were selected), and the exclusion of studies, results of which could not be transformed into OR and 95% CI, could be partly attributable to the publication bias. Since no significant heterogeneities were observed in all meta-analyses and the number of studies in each analysis is less than 5, sensitivity analysis is inappropriate for this meta-analysis.

Conclusions

In this meta-analysis, we found a clear association between different levels of PM_{2.5} exposure and preterm birth. The results highlight the needs to protect expectant mothers from PM_{2.5} exposures, especially during their first trimester of pregnancy even when the ambient PM_{2.5} concentration is relatively low. However, the limited number of studies restricts us from further exploring the relationship between PM_{2.5} exposed in different geographic areas and preterm birth, and we could not conduct sensitivity analysis to test the robustness of the results. Therefore, more prospective studies are needed in future meta-analyses to demonstrate that PM_{2.5} exposure is a risk factor of preterm birth, and more relevant health policy should be carried out to prevent that hazard effect.

Compliance with ethical standards

Funding The study was funded by the Hubei Province Health and Family Planning Scientific Research Project (WJ2015MA027) and the National Natural Science Foundation of China (no. 91543207).

Conflict of interest The authors declare that they have no conflict of interest.

References

- Bosetti C, Nieuwenhuijsen MJ et al (2010) Ambient particulate matter and preterm birth or birth weight: a review of the literature. *Arch Toxicol* 84(6):447–460
- Brauer M, Lencar C et al (2008) A cohort study of traffic-related air pollution impacts on birth outcomes. *Environ Health Perspect* 116(5):680–686
- Cha J, Bartos A et al (2013) Combinatory approaches prevent preterm birth profoundly exacerbated by gene-environment interactions. *J Clin Invest* 123(9):4063–4075
- Chen Y, Mi BB et al (2016) Prenatal exposure to outdoor air pollution and preterm birth: a Meta-analysis. *Chin J Epidemiol* 37(6):880–885
- Comnier SA, Ye X et al (2016) Acute effects of particulate air pollution on the incidence of coronary heart disease in Shanghai, China. *PLoS One* 11(3):e0151119
- DeFranco E, Moravec W et al (2016) Exposure to airborne particulate matter during pregnancy is associated with preterm birth: a population-based cohort study. *Environ Health* 15(1)

- DerSimonian R, Laird N (2015) Meta-analysis in clinical trials revisited. *Contemp Clin Trials* 45:139–145
- DeVries R, Kriebel D et al (2016) Low level air pollution and exacerbation of existing COPD: a case crossover analysis. *Environ Health* 15(1)
- Fleischer NL, Merilampi M et al (2014) Outdoor air pollution, preterm birth, and low birth weight: analysis of the World Health Organization global survey on maternal and perinatal health. *Environ Health Perspect* 122(4):425–151
- Gehring U, Wijga AH et al (2011) Traffic-related air pollution, preterm birth and term birth weight in the PIAMA birth cohort study. *Environ Res* 111(1):125–135
- Gent JF, Triche EW et al (2003) Association of low-level ozone and fine particles with respiratory symptoms in children with asthma. *JAMA* 290(14):1859–1867
- GK S (2012) Preterm birth, a known risk factor for infant and childhood death, is an independent risk factor for mortality in early childhood and young adulthood. *Evid Based Med* 17:121–122
- Glinianaia SV, Rankin J et al (2004) Particulate air pollution and fetal health. *Epidemiology* 15(1):36–45
- Ha S, Hu H et al (2014) The effects of air pollution on adverse birth outcomes. *Environ Res* 134:198–204
- Hannam K, McNamee R et al (2014) Air pollution exposure and adverse pregnancy outcomes in a large UK birth cohort: use of a novel spatio-temporal modelling technique. *Scand J Work Environ Health* 40(5):518–530
- Hatala R (2005) Tips for learners of evidence-based medicine: 4. Assessing heterogeneity of primary studies in systematic reviews and whether to combine their results. *Can Med Assoc J* 172(5):661–665
- Heo J, Schauer JJ et al (2014) Fine particle air pollution and mortality. *Epidemiology* 25(3):379–388
- Huttunen K, Siponen T et al (2012) Low-level exposure to ambient particulate matter is associated with systemic inflammation in ischemic heart disease patients. *Environ Res* 116:44–51
- Jalaludin B, Mannes T et al (2007) Impact of ambient air pollution on gestational age is modified by season in Sydney, Australia. *Environ Health* 6(1)
- Jedrychowski W, Bendkowska I et al (2004) Estimated risk for altered fetal growth resulting from exposure to fine particles during pregnancy: an epidemiologic prospective cohort study in Poland. *Environ Health Perspect* 112(14):1398–1402
- Jedrychowski W, Perera F et al (2009) Gender differences in fetal growth of newborns exposed prenatally to airborne fine particulate matter. *Environ Res* 109(4):447–456
- Kannan S, Misra DP, Dvonch JT, Krishnakumar P et al (2006) Exposures to airborne particulate matter and adverse perinatal outcomes: a biologically plausible mechanistic framework for exploring potential effect modification by nutrition. *Environ Health Perspect* 114(11):1636–1642
- Lamichhane DK, Leem J-H et al (2015) A meta-analysis of exposure to particulate matter and adverse birth outcomes. *Environ Health Toxicol* 30:e2015011
- Lee P-C, Roberts JM et al (2012) First trimester exposure to ambient air pollution, pregnancy complications and adverse birth outcomes in Allegheny County, PA. *Matern Child Health J* 17(3):545–555
- Liu J-P (2008) Design, implementation and methodological issues in cohort study. *J Chin Integr Med* 6(4):331–336
- Maisonet M, Correa A et al (2004) A review of the literature on the effects of ambient air pollution on fetal growth. *Environ Res* 95(1):106–115
- Matte T, Clougherty JE et al (2016) Ambient fine particulate matter, nitrogen dioxide, and preterm birth in New York City. *Environ Health Perspect* 124(8)
- Nieuwenhuijsen MJ, Dadvand P, Grellier J, Martinez D, Vrijheid M (2013) Environmental risk factors of pregnancy outcomes: a summary of recent meta-analyses of epidemiological studies. *Environ Health* 12(6)
- Pereira G, Belanger K et al (2013) Fine particulate matter and risk of preterm birth in Connecticut in 2000–2006: a longitudinal study. *Am J Epidemiol* 179(1):67–74
- Pereira G, Bell ML et al (2014) Fine particulate matter and risk of preterm birth and pre-labor rupture of membranes in Perth, Western Australia 1997–2007: a longitudinal study. *Environ Int* 73:143–149
- Pereira G, Evans KA et al (2016) Fine particulates, preterm birth, and membrane rupture in Rochester, NY. *Epidemiology* 27(1):66–73
- Poirier A, Dodds L et al (2015) Maternal exposure to air pollution and adverse birth outcomes in Halifax, Nova Scotia. *J Occup Environ Med* 57(12):1291–1298
- Qian Z-M, Liang S-W et al (2016a) Ambient air pollution and preterm birth: a prospective birth cohort study in Wuhan, China. *Int J Hyg Environ Health* 219(2):195–203
- Qian N-S, Yu H-T et al (2016b) Association between maternal air pollution exposure during first trimester and birth weight in Shanghai. *J Occup Environ Med* 33(9):827–832
- Ritz B, Wilhelm M et al (2007) Ambient air pollution and preterm birth in the environment and pregnancy outcomes study at the University of California, Los Angeles. *Am J Epidemiol* 166(9):1045–1052
- Romero R, Dey SK et al (2014a) Preterm labor: one syndrome, many causes. *Science* 345(6198):760–765
- Romero R, Miranda J et al (2014b) Prevalence and clinical significance of sterile intra-amniotic inflammation in patients with preterm labor and intact membranes. *Am J Reprod Immunol* 72(5):458–474
- Sapkota A, Chelikowsky AP et al (2010) Exposure to particulate matter and adverse birth outcomes: a comprehensive review and meta-analysis. *Air Qual Atmos Health* 5(4):369–381
- Shi L, Zanobetti A, Kloog I, Coull BA, Koutrakis P et al (2015) Low-Concentration PM_{2.5} and Mortality: Estimating Acute and Chronic Effects in a Population-Based Study. *Environ Health Perspect* 124(1):46–52
- Šrám RJ, Binková B et al (2005) Ambient air pollution and pregnancy outcomes: a review of the literature. *Environ Health Perspect* 113(4):375–382
- Stieb DM, Chen L, Eshoul M, Judek S (2012) Ambient air pollution, birth weight and preterm birth: a systematic review and meta-analysis. *Environ Res* 117:100–111
- Stroup DF (2000) Meta-analysis of observational studies in epidemiology: a proposal for reporting. *JAMA* 283(15):2008
- Sun X-L, Luo X-P et al (2015) The association between fine particulate matter exposure during pregnancy and preterm birth: a meta-analysis. *BMC Pregnancy Childbirth* 15(1)
- Wang J-J, Zhao A-L et al (2009) Impact of air pollution on low birth weight and preterm birth: a meta-analysis. *J Environ Health* 26(9):795–798
- Wegorzewska M, Nijagal A et al (2014) Fetal intervention increases maternal T cell awareness of the foreign conceptus and can lead to immune-mediated fetal demise. *J Immunol* 192(4):1938–1945
- Laurent O, Hu J, Li L, Kleeman MJ, Bartell SM et al (2016) A Statewide Nested Case–Control Study of Preterm Birth and Air Pollution by Source and Composition: California, 2001–2008. *Environ Health Perspect* 124(9):1479–1486
- Xu D-Z (2008) Research design of clinical medicine research. *Chin J Pract Pediatr* 23(3):237–240
- Zeng X-T, Liu H et al (2012) Meta-analysis: quality assessment tools of observational study. *Chin J Evid Based Cardiovasc Med* 4(4):297–299
- Zhang L, Zhang Y et al (2015) Characteristics of atmospheric PM_{2.5} and the variation of PAHs in PM_{2.5} during spring in Hongshan district, Wuhan. *China Environ Sci* 35(8):2319–2325
- Zhu X-X, Liu Y et al (2014) Maternal exposure to fine particulate matter (PM_{2.5}) and pregnancy outcomes: a meta-analysis. *Environ Sci Pollut Res* 22(5):3383–3396